



1. Identification of the substance/preparation and of the company undertaking

Trade Name: KPL Zinc Carbonate

Application: Used as a precursor in zinc oxide production, in rubber, ceramics, and as a dietary supplement or flame retardant additive

Manufacturer/Supplier:

Kesar Petroproducts Ltd.

Plot No. D-7/1, Lote Parshuram MIDC, Tal-Khed, Dist. Ratnagiri, Maharashtra – 415722

Further Information: Product Safety Department, Kesar Petroproducts Ltd., Ratnagiri Plant

2. Hazards Identification

GHS Classification:

Not classified as hazardous under EU CLP or OSHA HCS for standard industrial use

May cause respiratory irritation (H335) if fine dust is inhaled

Label Elements:

Pictogram: None required

Precautionary Statement: P261 – Avoid breathing dust; P271 – Use only in well-ventilated areas

3. Composition/Information on Ingredients

Description: Zinc Carbonate

Chemical Formula: ZnCO_3

CAS No.: 3486-35-9

Purity: $\geq 98\%$ (typical industrial grade)

Decomposition: Decomposes to ZnO and CO_2 at $\sim 300^\circ\text{C}$

4. First-Aid Measures

Inhalation: Move to fresh air. If symptoms (coughing, shortness of breath) occur, seek medical attention.

Skin Contact: Wash with soap and water.

Eye Contact: Rinse thoroughly with water for at least 15 minutes. Consult a physician if irritation persists.

Ingestion: Rinse mouth. Do NOT induce vomiting. Seek medical help if large quantity ingested.

5. Fire-Fighting Measures

Extinguishing Media: Water spray, dry chemical, CO_2 , or foam

Special Hazards: Decomposes on heating to release carbon dioxide and zinc oxide fumes (may cause metal fume fever)

Firefighter Protection: Wear full protective equipment and self-contained breathing apparatus

6. Accidental Release Measures

Personal Precautions: Wear dust mask (N95), safety goggles, and gloves

Environmental Precautions: Prevent entry into drains, soil, or waterways

Cleanup Methods: Dampen dust to suppress airborne particles. Collect in sealed container for disposal. Avoid dry sweeping.

7. Handling and Storage

Handling: Use in well-ventilated area. Avoid creating dust. Ground containers during transfer.

Storage: Store in a cool, dry place in tightly closed containers. Protect from moisture and acids.

8. Exposure Controls/Personal Protection

Occupational Exposure Limit:

Zinc (as Zn): ACGIH TLV-TWA 5 mg/m³ (respirable fraction)

Engineering Controls: Local exhaust ventilation at dust-generating operations

PPE:

Respiratory: NIOSH-approved particulate respirator if dust > permissible levels

Hand Protection: Nitrile or rubber gloves

Eye Protection: Safety goggles

Body Protection: Protective work clothing

9. Physical and Chemical Properties

Appearance: White, odorless powder

Odour: Odourless

pH (10% aqueous suspension): 6.5 – 7.5

Specific Gravity: ~4.4

Melting Point: Decomposes at ~300°C (does not melt)

Solubility in Water: Very low (<0.02 g/100 mL at 20°C)

Decomposition: $\text{ZnCO}_3 \rightarrow \text{ZnO} + \text{CO}_2$ (on heating)

10. Stability and Reactivity

Stable under normal storage and handling conditions

Incompatible with: Strong acids (releases CO₂ gas and zinc salts)

Hazardous Decomposition Products: Zinc oxide fumes (toxic if inhaled), carbon dioxide

11. Toxicological Information

Acute Toxicity (oral, rat): LD₅₀ > 2,000 mg/kg (low toxicity)

Inhalation: Low acute risk; chronic high-dose dust exposure may cause respiratory irritation

Skin/Eye: Not a significant irritant; mechanical irritation possible from dust

Carcinogenicity: Not classified as carcinogenic (IARC Group 3)

12. Ecological Information

Aquatic Toxicity: Zinc ions released may be harmful to aquatic organisms

Persistence: Zinc may persist in sediments

Precaution: Do not discharge into environment without proper treatment

13. Disposal Considerations

Dispose in accordance with local, state, and national regulations

Due to zinc content, may be subject to hazardous waste rules in some regions—verify with local authority

Packaging: Rinse and dispose as industrial waste or recycle per local guidelines

14. Transportation Information

UN Number: Not regulated

Proper Shipping Name: Zinc carbonate

Transport Classification: Not restricted under IMDG, IATA, ADR, or RID

15. Regulatory Information

GHS Classification: Not classified as hazardous

REACH: Registered (if tonnage >1 tpa)

TSCA: Listed

16. Other Information

This information is compiled in good faith based on current scientific knowledge. It is not a substitute for professional judgment. Kesar Petroproducts Ltd. assumes no liability for use beyond recommended conditions.

Issued by: Quality Control Department, Kesar Petroproducts Ltd.

Date: 29 December 2025